RES-5 endpoint 2PI bound: the endpoint marginalit-y is an artefact of the

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

res5-tail-budget-closure localised RES-5 to the $I = 2 \times 10^{-3}$ endpoint, where the operator-norm tail $C_G a_0 = 0.047$ exceeds the conservative slack 0.0385 (tail/slack = 1.22). This note resolves the endpoint by bracketing it between the two SC-SCOPE floor grades and identifying the single rigorous residual. Scope: the $I = 2 \times 10^{-3}$ endpoint at the operating anchor; estimate/verified grade as marked. No tier action.

2. The slack: proved vs verified floor

The third-cumulant slack $1 - 1/\text{joint}$ is set by the SC-SCOPE endpoint floor ρ_{lat}, for which two grades are on record (scscope-quartic-normalisation-certificate):

$$\begin{aligned} \text{PROVED } (K_{\text{floor}} \le T') : &\quad \rho_{\text{lat}} = 6.55, \quad \text{joint} = 1.040, \quad \text{slack} = 0.0385, \\ \text{VERIFIED } (K_{\text{floor}} \le 0.52\, T') : &\quad \rho_{\text{lat}} = 12.6, \quad \text{joint} = 1.082, \quad \text{slack} = 0.0758. \end{aligned}$$

The inequality $K_{\text{floor}} \le T'$ is a proved theorem (Parseval + Lemma A); the factor 0.52 is the numerically realized value (verified, not yet proved).

3. The tail: operator-norm vs projection

The endpoint tail estimate $C_{\text{higher}}/\Delta F_{\text{margin}} \approx C_G a_0$, $C_G = 1/(1+g) = 0.492$, $a_0 = 0.096$, is an OPERATOR-NORM upper bound: it uses the full response norm $\|(1 + G K_{\text{BS}} G)^{-1}\| = 1/(1+g)$. The actual pattern-dependent free-energy difference carries a projection factor

$$\frac{C_{\text{higher}}}{\Delta F_{\text{margin}}} = \chi_{\text{proj}}\, C_G a_0, \qquad \chi_{\text{proj}} \le 1,$$

because the source $\lambda'(P^2 - \langle P^2 \rangle)$ is common-mode-subtracted (R-U10-3) and supported on the $\{110\}$ reciprocal shell, which is NOT the softest screened mode that saturates the operator norm. Hence $\chi_{\text{proj}} < 1$ strictly.

4. Verdict: the endpoint is bracketed; closure at the thin-certified grade

Putting the two together, the operator-norm tail lies BETWEEN the proved and verified slacks:

$$\boxed{\text{slack}_{\text{proved}}(0.0385) \;<\; \text{tail}\,(0.047) \;<\; \text{slack}_{\text{verified}}(0.0758)}$$

Two independent, physically motivated, numerically verified sharpenings each close the endpoint:

(i) **Verified floor.** With the realized $K_{\text{floor}} \le 0.52T'$ ($\rho_{\text{lat}} = 12.6$, slack 0.0758) the endpoint closes with a 38% margin (tail/slack = 0.62).

(ii) **Projection.** At the proved slack 0.0385 the endpoint closes whenever $\chi_{\text{proj}} < 0.0385/0.047 = 0.82$.

The endpoint marginalit-y (tail/slack = 1.22) is therefore an ARTEFACT of simultaneously taking the most conservative floor AND the operator-norm tail.
Unification. Route (i) is the SAME inequality as SC-SCOPE's floor sharpening, $K_{\text{floor}} \le 0.52T'$. RES-5's last analytic piece and SC-SCOPE's named hypothesis collapse to ONE residual, at the SAME thin-certified grade: the higher-skeleton tail does not weaken B1 below the already-accepted SC-SCOPE thin-certified lift – it matches it.
Tier. RES-5/GAP-2: ENDPOINT-LOCALISED → STRONG EVIDENCE (closed off-endpoint at $\ge 27\times$ margin; closed at the endpoint at the SC-SCOPE thin-certified grade). No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}; SC-SCOPE thin-certified remains the named hypothesis until the $0.52T'$ floor is proved.

5. Estimate vs theorem (CLAUDE.md 6.3.5b)

The endpoint closure is STRONG EVIDENCE, not a theorem. $\text{slack}_{\text{verified}}$ rests on the verified (not proved) factor 0.52; $\chi_{\text{proj}} < 1$ is physically established but not yet bounded below 0.82. The PROVED-grade endpoint ($K_{\text{floor}} \le T'$, operator-norm tail) remains marginal. The single rigorous residual is therefore: prove $K_{\text{floor}} \le 0.52T'$ (which also discharges SC-SCOPE) OR prove $\chi_{\text{proj}} \le 0.82$. Either closes RES-5 at the endpoint as a T6 result.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"Using the verified floor instead of the proved floor is choosing the favourable estimate."* VALID-with-mitigation: explicitly graded – the verified floor gives STRONG EVIDENCE, not a theorem; the proved-grade endpoint is stated as still marginal. The verified 0.52 is the realized value, not an optimistic guess; the rigorous gap is named (prove $0.52T'$).

(β) *"$\chi_{\text{proj}} < 1$ is asserted from support, not computed."* VALID: $\chi_{\text{proj}} < 1$ is a strict consequence of the common-mode subtraction (the constant mode is removed) and the shell support; the QUANTITATIVE $\chi_{\text{proj}} < 0.82$ is the registered residual, not claimed here.

(γ) *"This does not prove the endpoint – it relabels open as strong-evidence."* PARTLY-UPHELD: correct that no theorem is proved; the advance is (1) showing the marginalit-y is an artefact of the over-conservative floor, not a robust obstruction, and (2) unifying RES-5's residual with SC-SCOPE's, so closing ONE inequality discharges BOTH. The honest tier is STRONG EVIDENCE; B1 is unchanged.

Quantitative sanity check (mandatory): *bracket* – $0.0385 < 0.047 < 0.0758$; *verified margin* – $1 - 0.047/0.0758 = 38\%$; *projection threshold* – $0.0385/0.047 = 0.82 < 1$; *consistency* – ρ_{lat} doubling $6.55 \to 12.6$ raises the slack $0.0385 \to 0.0758$ ($\times 1.97$), matching joint $1.040 \to 1.082$.

7. Result footer

Result ID:	B1-RH-ENUM / RES-5 endpoint 2PI bound
Precise statement:	At the I=2e-3 endpoint the operator-norm tail C_G a0=0.047 is BRACKETED: slack_proved(0.0385) < tail < slack_verified (0.0758). With the realized verified floor K_floor<=0.52T' (rho_lat=12.6) the endpoint closes (tail/slack=0.62, 38% margin); equivalently it closes at the proved slack for chi_proj<0.82. Route (i) IS the SC-SCOPE floor sharpening: RES-5 and SC-SCOPE share one residual at one (thin-certified) grade. Marginalit-y is an artefact of the conservative floor.
Scope:	I=2e-3 endpoint, operating anchor; STRONG EVIDENCE / verified grade as marked. Proved-grade endpoint stays marginal.
Dependencies:	res5-tail-budget-closure (localisation); scscope-quartic-normalisation-certificate (joint 1.040/1.082, floor grades); res5-a0-skeleton-sensitivity (C_G, a0); R-U10-3 (common-mode subtraction).
Evidence grade:	ANALYTIC (bracket + projection) + EXECUTED (res5_endpoint_2pi_bound.py 5/5) + INHERITED (SC-SCOPE floor).
Reproduction command:	python codes/vacuum/res5_endpoint_2pi_bound.py
Expected output:	tail 0.047 in [0.0385, 0.0758]; tail/slack_verified 0.62; chi_proj threshold 0.82; 5/5 PASS.
Falsification gate:	A proof that chi_proj > 0.82 AND K_floor > 0.52T' (endpoint robustly open); or the verified floor failing at the endpoint.
Tier before / after:	No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2: ENDPOINT-LOCALISED -> STRONG EVIDENCE (endpoint closed at the SC-SCOPE thin-certified grade; rigorous T6 pending one inequality).
No-overclaim statement:	No theorem is proved. The endpoint closes at STRONG EVIDENCE (verified floor / projection), not rigorously; the proved-grade endpoint remains marginal. The advance is (a) the marginalit-y is an artefact of the conservative floor and (b) RES-5's residual unifies with SC-SCOPE's. B1 unchanged.
Next required action:	Prove K_floor <= 0.52 T' at the I=2e-3 endpoint (discharges BOTH RES-5 endpoint and SC-SCOPE), OR prove chi_proj <= 0.82 (the {110}-shell projection of the screened response). Either upgrades RES-5 endpoint to T6.